CO5- ASSESSMENT TOOL 2
Technical Presentation

CSA1709 - Artificial Intelligence

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Questions

1. System Analysis Presentation

Present a reverse engineering analysis of an AI-based system, explaining its architecture, components, and working process using diagrams.

2. Architecture Reconstruction Demo

Prepare a technical presentation to reconstruct the architecture of a given software/AI system, highlighting module interactions and data flow.

3. Algorithm & Logic Identification

Present how you identified algorithms and decision-making logic in a reverse-engineered system, including methodology and tools used.

4. Data Flow & Processing Pipeline

Explain through a presentation the data flow, processing pipeline, and communication mechanisms in a reverse-engineered distributed system.

5. Enhancement & Redesign Proposal

Deliver a technical presentation proposing improvements or redesign strategies for an existing system based on reverse engineering findings.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

1.SYSTEM ANALYSIS PRESENTATION

Introduction & Problem Identification

- Reverse engineering analyzes an existing AI system.
- It helps understand:
  - Architecture
  - Components
  - Algorithms
  - Data flow
  - Working process
- The original functionality is not modified.
- Main problem: internal components and dependencies may be undocumented.
- Reverse engineering helps identify bottlenecks and improvement areas.

System Architecture

Flow:

User / Input → Data Collection → Data Preprocessing → AI/ML Processing → Decision-Making → Output

Main Layers:

- Input Layer – collects data
- Data Processing Layer – cleans and transforms data
- AI Processing Layer – performs AI/ML operations
- Decision Layer – generates decisions/predictions
- Output Layer – communicates the result
- Storage Layer – stores data, models and results

Working Process & Tools

Working Process

- Receive input from user/external source.
- Validate and preprocess the data.
- Send processed data to AI model.
- Perform prediction or decision-making.
- Interpret the result.
- Return final output.

Tools Used:

- Python
- PySpark
- RDD operations
- System monitoring tools
- Architecture diagrams
- Log analysis

2.ARCHITECTURE RECONSTRUCTION DEMO

Objective & Problem

Objective:

- Reconstruct an existing AI system.
- Identify individual modules.
- Understand module interactions.
- Represent complete data flow.

Problems:

- Undocumented modules may exist.
- Dependencies may be difficult to identify.
- Complete architecture may not be directly visible.

Reconstructed Architecture

Flow

User/Input ➡ Data Collection ➡ Preprocessing ➡ AI Processing ➡ Decision Module ➡ Output / Result

Module Interaction:

- Input sends raw data to preprocessing.
- Preprocessing converts raw data.
- AI module processes prepared data.
- Decision module interprets AI results.
- Output module provides final response.

Data Flow

Data Flow Through the System

Raw Data
↓
Cleaning
↓
Transformation
↓
Feature/Data Representation
↓
AI Processing
↓
Decision
↓
Final Output

Supporting Tools:

- Architecture diagrams
- Python
- PySpark
- Log analysis
- Performance monitoring
- Data-flow visualization

3.ALGORITHM & LOGIC IDENTIFICATION

Objective & Methodology

Objectives:

- Identify algorithms used in the AI system.
- Understand how decisions are made.
- Connect input, processing logic and output.

Methodology:

- Input Analysis
- Data-Flow Analysis
- Code/Logic Analysis
- Algorithm Identification
- Output Analysis

Algorithm Identification

Input Analysis

- Identify input type and format.
- Understand how data enters the system.

Data-Flow Analysis

- Trace data through modules.
- Identify transformations.

Code/Logic Analysis

- Examine functions and modules.
- Identify loops, conditions and mathematical operations.

Algorithm Identification

- Compare processing patterns with known algorithms.
- Identify AI/ML algorithms based on operations and outputs.

Decision-Making Flow

Decision-Making Process

Input Data
↓
Preprocessing
↓
Feature/Data Extraction
↓
Algorithm
↓
Prediction / Score
↓
Decision Rules
↓
Final Output

Tools:

- Python
- PySpark
- RDD operations
- Debuggers
- Log analysis
- Code analysis
- Performance monitoring

4.DATA FLOW & PROCESSING PIPELINE

Introduction & Problem

Data Flow and Processing Pipeline in a Distributed AI System

- Distributed AI systems process large datasets using multiple computing resources.
- Data is divided and processed across different components.
- Results are transferred and combined.
- Large datasets can make centralized processing slow.
- Communication can introduce delays.
- Fault tolerance and data consistency are important.

Distributed Processing Pipeline

Processing Pipeline

Data Sources

↓

Data Ingestion

↓

Data Partitioning

↓

Distributed Processing

↓

AI/ML Computation

↓

Aggregation

↓

Result Storage

↓

User / Application

Important Factors & Tools

Important Factors:

- Data partitioning
- Parallel processing
- Network communication
- Fault tolerance
- Resource utilization
- Result aggregation

Tools:

- PySpark
- RDD , Logs
- Spark monitoring tools
- Performance monitoring
- Data-flow diagrams

5.ENHANCEMENT & REDESIGN PROPOSAL

Objectives:

- Identify limitations found during reverse engineering.
- Propose suitable technical improvements.
- Redesign the system for:
  - Better performance
  - Higher scalability
  - Improved reliability
  - Easier maintenance

Proposed Improvements

Existing Problem	Proposed Enhancement
Slow data processing	Parallel distributed processing
Large data volume	PySpark/RDD-based processing
High communication overhead	Optimize data transfer
Poor scalability	Scalable architecture
Difficult debugging	Better logging and monitoring
Resource wastage	Optimize CPU and memory
Single point of failure	Fault-tolerant mechanisms

Benefits & Ethics

Benefits:

- Faster processing.
- Better scalability.
- Improved CPU and memory utilization.
- Higher system reliability.
- Easier maintenance.
- Better monitoring and debugging.
- Improved handling of large datasets.

Ethical & Professional Considerations:

- Protect sensitive data.
- Maintain transparency in AI decisions.
- Validate outputs before deployment.
- Avoid unauthorized system modification.
- Document reverse-engineering findings and redesign decisions.

Thank

You….